

WORKSHEET - NAPLAN Year 9

Number and Algebra Basic Calculations and Algebra (9)

Teacher: Joanne Rust

Exam Equivalent Time: 10 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. Number and Algebra, NAP-85047

A return trip from Brodie's house to the beach is 5.78 kilometres.

How far does Brodie travel if he does this 14 times?

8.09 km 23.12 km 42.12 km 69.36 km 80.92 km

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2. Number and Algebra, NAP-08568

In Australia, 288 701 children were enrolled in kindergarten in 2013.

Of these children, 150 125 were boys.

How many girls were enrolled in kindergarten in 2013?

122 896 138 576 288 701 438 826

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3. Number and Algebra, NAP-61526

The airforce buys 8 new fighter jets for \$882.0 million.

Each jet costs the same amount.

What is the cost of 1 fighter jet?

\$110.25 million \$441.0 million \$874.0 million \$7056 million

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4. Number and Algebra, NAP-08598

30^2 is equal to which of the following?

$60^2 \div 3$ $3^2 \times 2 \times 5 \times 2 \times 5$ 9×5^2 $3 \times 10 \times 10$

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5. Number and Algebra, NAP-49699

What is the value of $\frac{74.2 + 13.56}{6.4 \times 2.1}$

4.6 6.5 10.3 74.9

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6. Number and Algebra, NAP-26208

The dog population in five continents can be found in the table below.

Continent	Dog population
Africa	9, 721, 457
Asia/Oceania	72, 428, 799
Europe	59, 329, 647
North America	43, 520, 201
South America	13, 262, 582

How many **more** dogs live in Europe than in Africa and South America combined?

7. Number and Algebra, NAP-38066

$$1200 = 1500 - \frac{a}{6}$$

What is the value of a ?

200 1200 1800 7200 16200

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8.  Number and Algebra, NAP-55674

$$120 - \frac{\sqrt{65^2 + 72^2}}{2} = \boxed{}$$

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Worked Solutions

1. Number and Algebra, NAP-85047

$$\begin{aligned}\text{Distance} &= 14 \times 5.78 \\ &= 80.92 \text{ km}\end{aligned}$$

2. Number and Algebra, NAP-08568

$$\text{Number of girls} = 288\,701 - 150\,125 = 138\,576$$

3. Number and Algebra, NAP-61526

$$\frac{\$882.0 \text{ million}}{8} = \$110.25 \text{ million}$$

4. Number and Algebra, NAP-08598

$$30^2 = 900$$

$$\begin{aligned}3^2 \times 2 \times 5 \times 2 \times 5 \\ &= 3^2 \times 10 \times 10 \\ &= 900\end{aligned}$$

5. Number and Algebra, NAP-49699

$$\begin{aligned}\frac{87.76}{13.44} &= 6.529\dots \\ &= 6.5\end{aligned}$$

6. Number and Algebra, NAP-26208

$$\begin{aligned}\text{Number of extra dogs} \\ &= 59\,329\,647 - (9\,721\,457 + 13\,262\,582) \\ &= 36\,345\,608\end{aligned}$$

7. Number and Algebra, NAP-38066

$$1200 = 1500 - \frac{a}{6}$$

$$\frac{a}{6} = 300$$

$$\therefore a = 1800$$

8. Number and Algebra, NAP-55674

$$120 - \frac{\sqrt{65^2 + 72^2}}{2}$$

$$= 120 - \frac{97}{2}$$

$$= 120 - 48.5$$

$$= 71.5$$